

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows the structure of a lysosome.

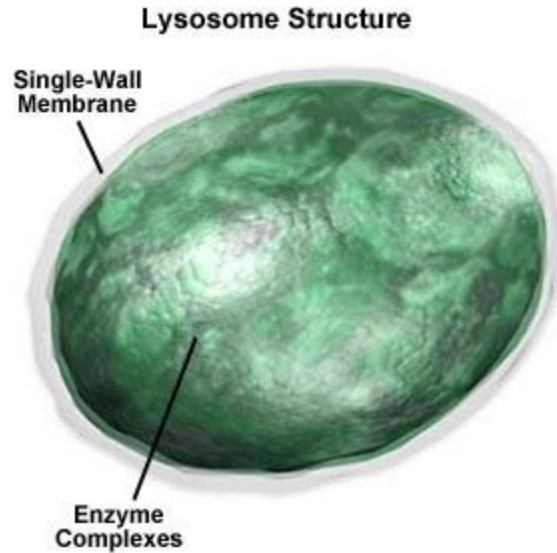


Fig. 1.1

- (a) State the main functions of the lysosome. [2]

- Release enzymes outside of the cell (**exocytosis**) which may serve the purpose of destroying materials around the cell.
- Digestion of materials from inside the cell (**autophagy**) i.e. by fusing with vacuoles from inside the cell. This could include digesting worn-out organelles so that useful chemicals locked-up in their structures can be re-used by the cell.
- Recycle the products of biochemical reactions that have taken place following materials being brought into the cell by endocytosis (general term for this 'recycling' function: **biosynthesis**).
- Completely break-down cells that have undergone apoptosis (**autolysis**)

[Max 2]

- (b) (i) With reference to Fig. 1.1, outline how the enzyme hexosaminidase A in the lysosome is formed. [4]

- Gene encoding for hexosaminidase A is transcribed in the nucleus;
- mRNA is then translated at bound ribosomes on the rough endoplasmic reticulum;
- polypeptide is then transported to the Golgi Apparatus for modification;
- Modified protein is then packaged and sorted into lysosome;

(ii) Suggest **two** reasons for insufficient activity of the enzyme hexosaminidase A. [2]

- Mutation in the DNA coding sequence, resulting in a change in the three-dimensional conformation of the enzyme, hence altering the conformation of the active site;
- Change in pH of the lysosome, not optimum pH;
- AVP;

(c) Explain how Tay-Sachs disease can affect nervous impulse transmission. [2]

- Loss of myelin sheath,
- Action potential propagation will be affected due to leaking of sodium and potassium ions.

[Total: 10]

2 Fig. 2.1 shows the formation of collagen fiber.

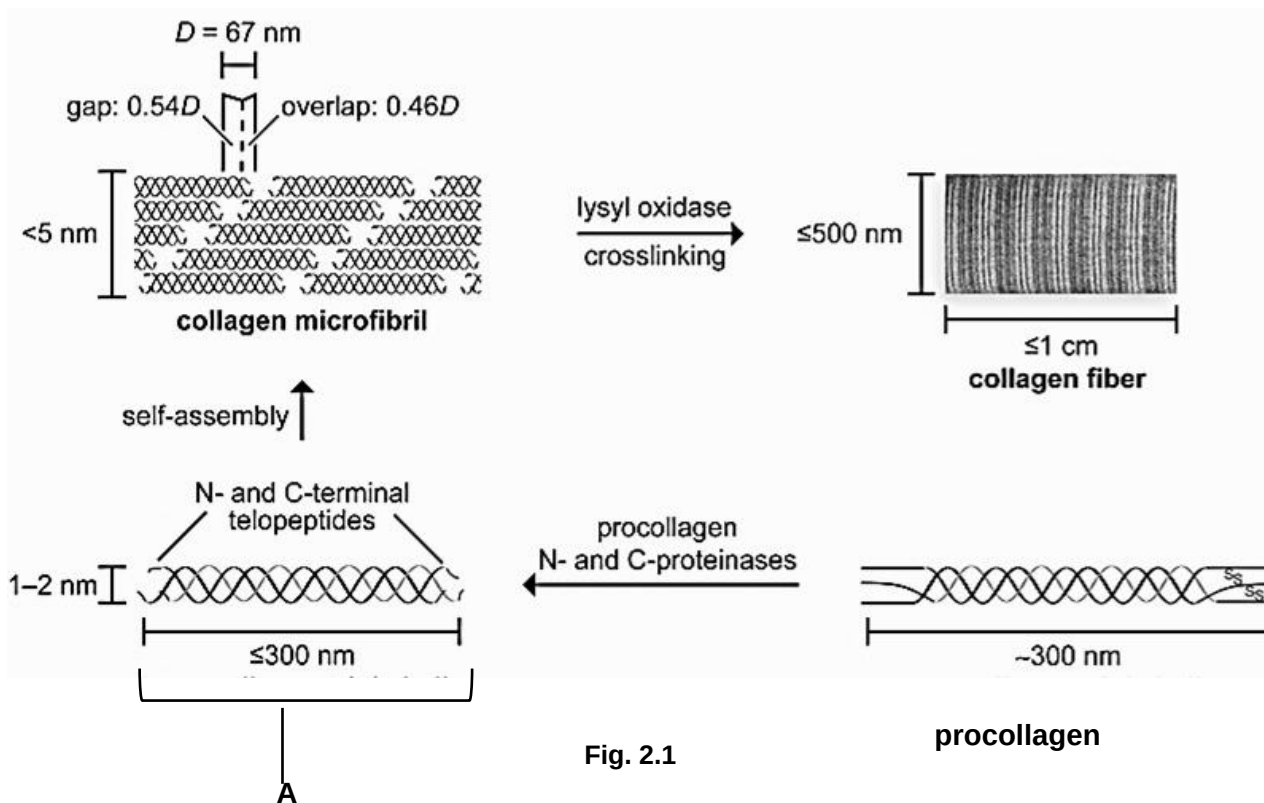


Fig. 2.1

(a) With reference to Fig. 2.1,

(i) describe structure A. [2]

- **Three helical polypeptides**, each wound around one another in a right-handed triple helix called **triplecollagen**.
- Is 300-nm long, thin (1-2nm diameter)
- Each polypeptide contains a **repetitive tripeptide sequence Glycine-X-Y**, where X is often proline and Y is often hydroxyproline or hydroxylysine.

[Max 2]

(ii) explain how the three chains in **A** are held together. [1]

- **(Interchain) hydrogen bonds** are formed between the **NH groups of glycine residues** on one chain and **C=O groups of proline residues / C=O groups of another amino acid residue** on an **adjacent chain** stabilises the tropocollagen structure / the three chains in A.

Modifications of the procollagen include hydroxylation, glycosylation, and disulfide bond formation. Hydroxylysines are glycosylated by the addition of galactose and elongated by the addition of glucose and galactose units.

(iii) Distinguish **two** main structural features in the galactose-glucose–galactose chain and a single chain found in procollagen. [2]

- **Type of monomers:** Amino acids in procollagen vs monosaccharides (glucose and galactose) in galactose-glucose–galactose chain
- **Type of bonds present between adjacent monomers:** Peptide bonds in procollagen vs glycosidic bonds galactose-glucose–galactose chain, formed via the process of condensation

(iv) Explain the need for disulfide bond formation in the modified procollagen. [1]

- Interchain disulfide bonds between the C-terminal propeptides of procollagen align the three chains together and helps initiate formation of the triple helix.

(v) State precisely the location in the cell where modifications to procollagen occur. [1]

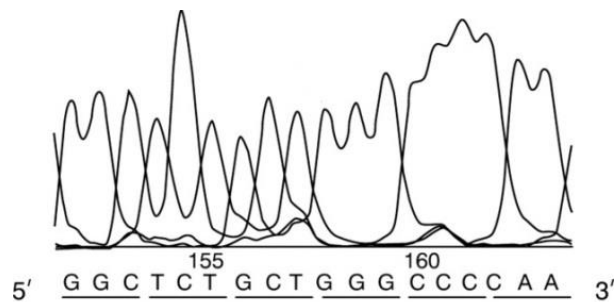
- In endoplasmic reticulum / Golgi apparatus

Osteogenesis imperfecta (OI) is a heritable connective tissue disorder characterised by varying degrees of bone fragility. OI is caused by mutations in one of the two α chains of type I collagen, the principal component of the extracellular matrix in bone and skin.

OI was found in a 9-year-old Turkish boy, whose parents are healthy and have no history of fractures. The boy exhibited a high fracture rate and his limb deformities have progressed to include a severely deformed pelvis, thin and markedly deformed long bones.

Sequencing of the normal and mutant DNA coding alleles was performed and the results are shown in Fig. 2.2.

Normal allele:



Mutant allele:

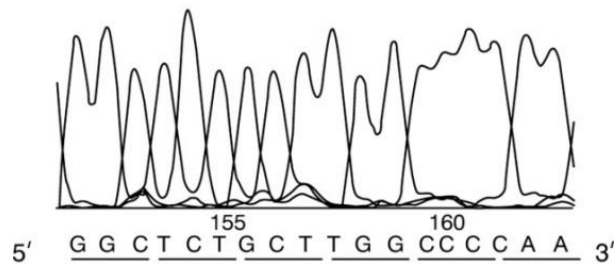


Fig. 2.2

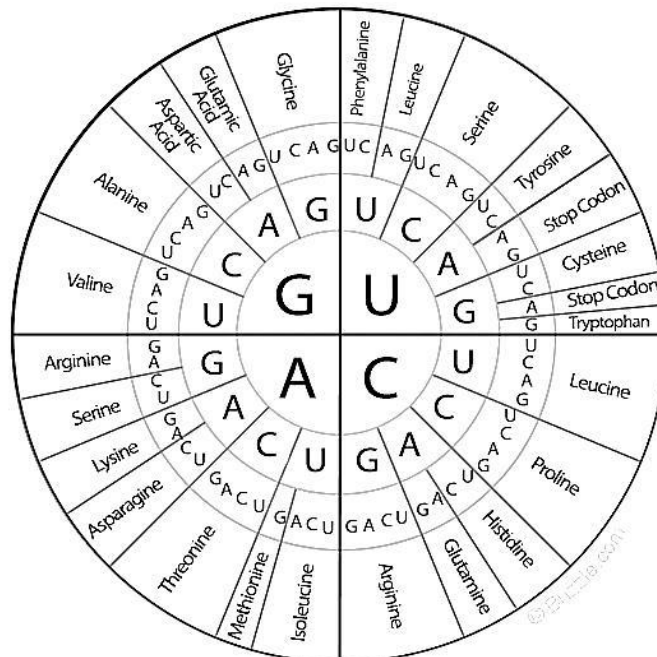


Fig. 2.3

(b) Using the information in Fig. 2.2 and the genetic code in Fig. 2.3, explain why the boy [3]
has a high fracture rate.

- The **first base** of the **DNA triplet code** in the **non-coding DNA strand** (that reads in the **3' → 5' direction**) **C** (in the normal allele) has been **substituted with A** (in the mutant allele) results in a change in the **codon** from **GGG to UGG** in the mRNA transcribed;
- As a result, the amino acid **glycine** which has a **small side chain**, is now **substituted with the amino acid tryptophan**, which has a **bulky side chain**;
- This causes a **change in the three-dimensional conformation** of the α -chain of type I collagen, as the **bulky side chain / R group of tryptophan** prevents the **tight coiling of the α -chain**, destabilising the triple helix structure in collagen.

[Total: 10]

- 3 (a) Distinguish between the reproductive cycles of the influenza virus and the Human Immunodeficiency Virus (HIV).

[4]

Stage	Influenza virus	HIV
Penetration (max 1)	Virus is taken in by <u>receptor-mediated endocytosis</u>	HIV envelope <u>fuses</u> with the host cell membrane envelope
Replication of viral nucleic acids and integration (max 2)	<u>Viral RNA dependent-RNA polymerase</u> copies the (-) sense RNA template into complementary (+) sense RNAs, which are templates for synthesis of (-) viral RNAs (viral genomes) <u>No integration</u> of viral nucleic acid into host cell genome;	- Reverse transcriptase reverse transcribes (+) sense viral RNA into complementary DNA strand, which is then used as a template for synthesis of the other complementary DNA strand, forming a <u>double-stranded cDNA molecule</u>. <u>Integrase</u> catalyses the integration of viral DNA into host genome, forming <u>provirus</u>
Assembly and release (max 2)	8 segmented viral RNAs <u>buds off</u> from the host cell with the membrane containing viral glycoproteins <u>Neuraminidase (NA)</u> cleaves sialic acid residues on the viral envelope to prevent newly formed virions clumping together.	- Copies of HIV proteins and HIV's RNA genome assemble near the host cell membrane to form a new virus particle - Newly assembled immature <u>HIV buds off</u> from the <u>infected macrophage / T lymphocytes</u>, taking the host membrane with it. - The <u>HIV protease</u> cleaves the long chains of <u>polyproteins</u> (REJECT: reference to polypeptide) into smaller functional proteins, forming a <u>mature HIV particle</u>.
All answers must be in complete sentences.		

SAM domain and HD domain-containing protein 1 (SAMHD1) is a protein that in humans, is encoded by the *SAMHD1* gene. SAMHD1 is a cellular enzyme, responsible for blocking replication of HIV in macrophages. It is an enzyme that exhibits phosphohydrolase activity, converting nucleotide triphosphates (NTPs) to triphosphate and a nucleoside (i.e. nucleotides without a phosphate group).

Fig. 3.1 shows the action of SAMHD1 and other cellular enzymes that target nucleic acids, on HIV replication.

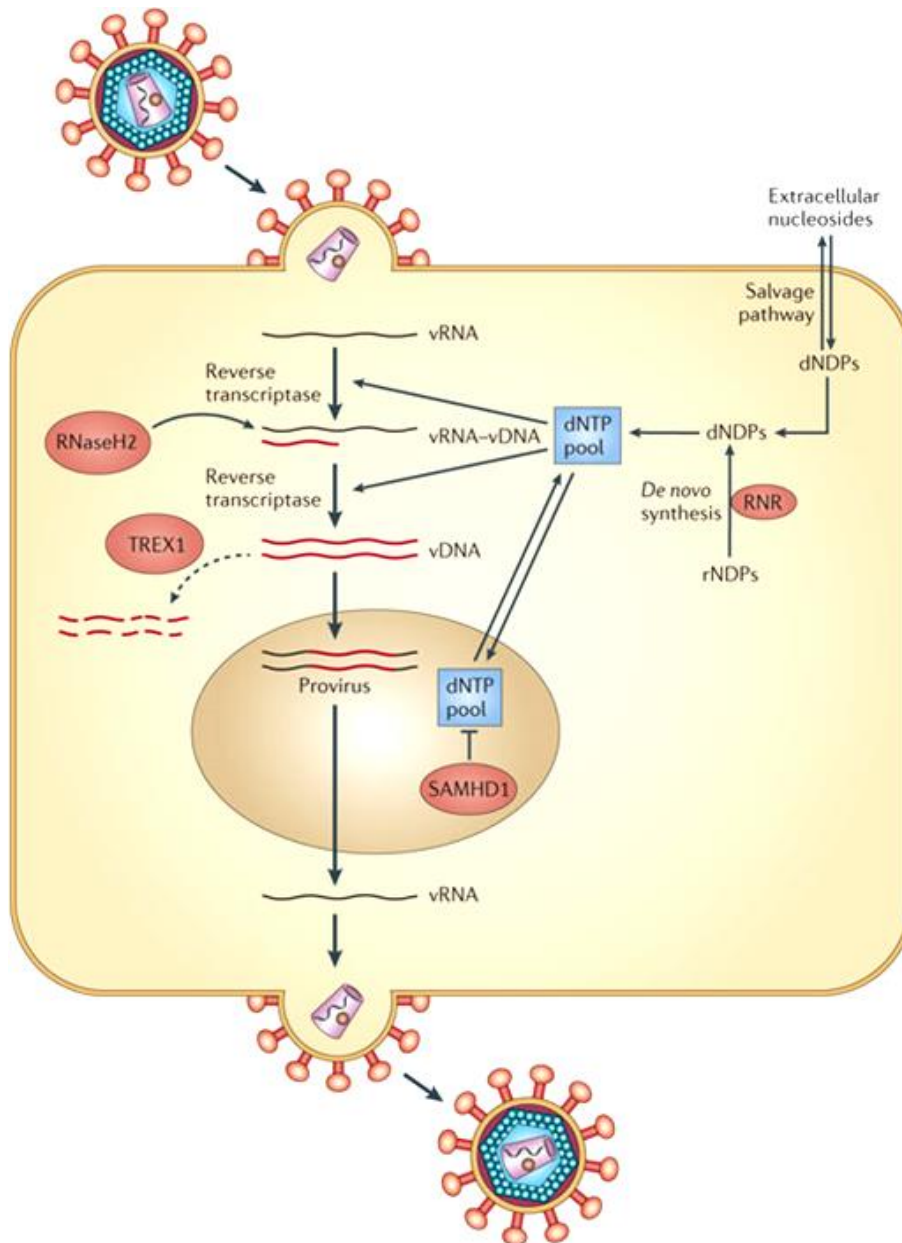


Image from *Nature Reviews Microbiology* **10**, 675-680 (2012)

Fig. 3.1

- (b) Using the information provided in the preamble and Fig. 3.1, explain how the action of SAMHD1 reduces the infectivity of HIV. [3]

- SAMHD1 **hydrolyses nucleotide triphosphates (NTPs)** to triphosphates and a nucleosides / hydrolyses deoxyribonucleotide triphosphates to triphosphates and deoxyribonucleosides.
- reducing the pool of NTPs available for reverse transcriptase** to carry out reverse transcription of single-stranded viral RNA to form double-stranded cDNA / viral DNA;

OR

Reducing the pool of NTPs required for formation of phosphodiester bonds between adjacent deoxyribonucleotides in viral DNA synthesis.

- resulting in **less viral DNA available for integration into macrophage cell's genome as a provirus** → reduction in latent cells;

Fig. 3.2 shows the changes in HIV proteins concentration and CD4 T-cells count in the bloodstream of an individual after HIV infection.

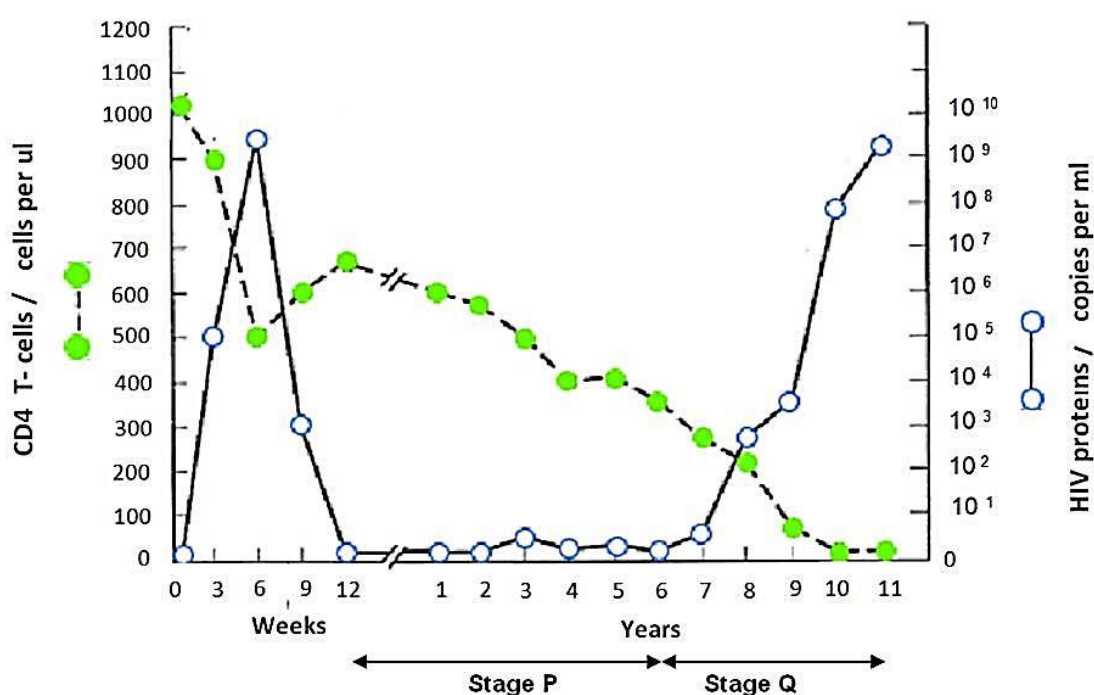


Fig. 3.2

- (c) (i) In an experiment, it was found that the concentration of the enzyme, histone deacetylase (HDAC) in infected cells at Stage P was higher than in infected cells not at Stage P.

Suggest an explanation for the presence of abnormally high levels of HDAC in infected cells at Stage P. [2]

- Histone deacetylase cause chromatin to be **more condensed/ compacted / packed more tighter**;

OR

Removal of acetyl groups from histones increases electrostatic attraction between histones and proviral DNA, making chromatin more condensed;

- Prevents transcription and translation of **viral proteins** during stage P;
- Viral proteins not present on infected cells → not detected and removed by immune system;

[any 2]

(ii) With reference to Fig. 3.2, describe and explain the changes in the amount of HIV proteins and CD4 T-cells count during Stage Q. [2]

- **(Describe)** From 6 to 11 years after infection, [viral proteins] increases from 0 to $10^9/10^{10}$ copies per ml **(Explain)** because viral proteins are actively expressed from integrated HIV DNA / provirus to form new viruses;
- **(Describe)** CD4 T cells decrease from 350 cells per μl to 0 cells per μl **(Explain)** as budding of large amount of virions from the cell surface of CD4 T cells may disrupt the cell membrane sufficiently for cell to die

OR

Hijacking of cellular machinery and resources towards producing new virions disrupts normal activities needed for cell survival, eventually causing cell death;

[Total: 11]

- 4 Cholesterol is taken up by liver cells, which transforms it into 30 to 100 nm low-density lipoprotein (LDL) particles. Fig. 4.1 shows the structure of an LDL particle. It consists of a core of triglycerides and a surface coat of phospholipids, cholesterol and a monomeric protein, ApoB-100.

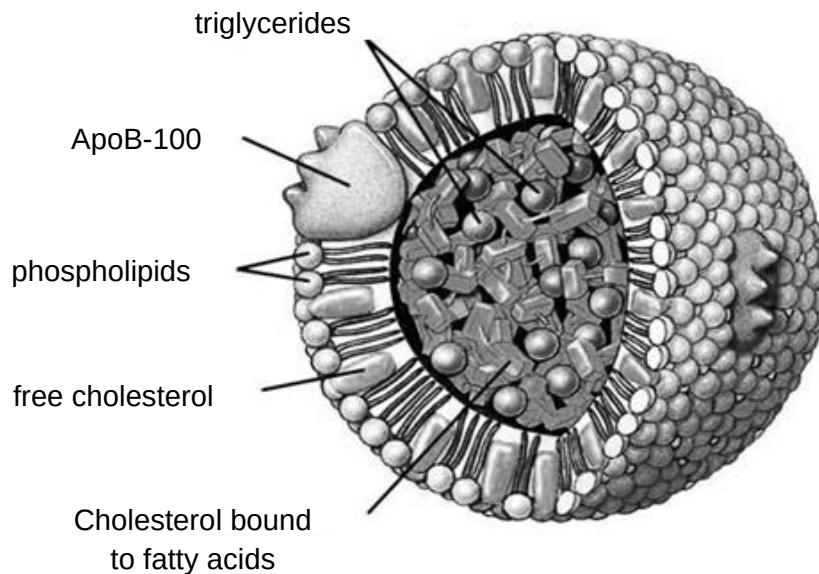


Fig. 4.1

- (a) With reference to Fig. 4.1, outline the roles of lipids in the LDL coat.

[3]

Cholesterol and phospholipids are important components of the LDL coat:

- Free cholesterol – maintains fluidity of the LDL coat.
- It disturbs the close packing of phospholipids thus keeping them more fluid.
- Phospholipids – major constituent of the LDL coat, they form the monolayer in LDL coat.
- The role of the phospholipid monolayer is to form a boundary between the hydrophobic triglyceride core from the aqueous exterior of the LDL.
- The hydrophobic fatty acid tails face inwards and make contact with the hydrophobic triglyceride core while the hydrophilic phosphate heads face outwards and make contact with the exterior aqueous environment.

A genetic form of extremely high levels of LDL cholesterol known as familial hypercholesterolaemia, leads to severe coronary artery disease and, often, premature death. A gene and its protein product, proprotein convertase subtilisin kexin type 9, or *PCSK9*, has been revealed to play a role in regulating LDL cholesterol levels in the blood. Patients with hypercholesterolaemia had a mutation in the *PCSK9* gene which leads to increased production of *PCSK9*.

Fig. 4.2 shows the regulatory pathway which *PCSK9* participates in.

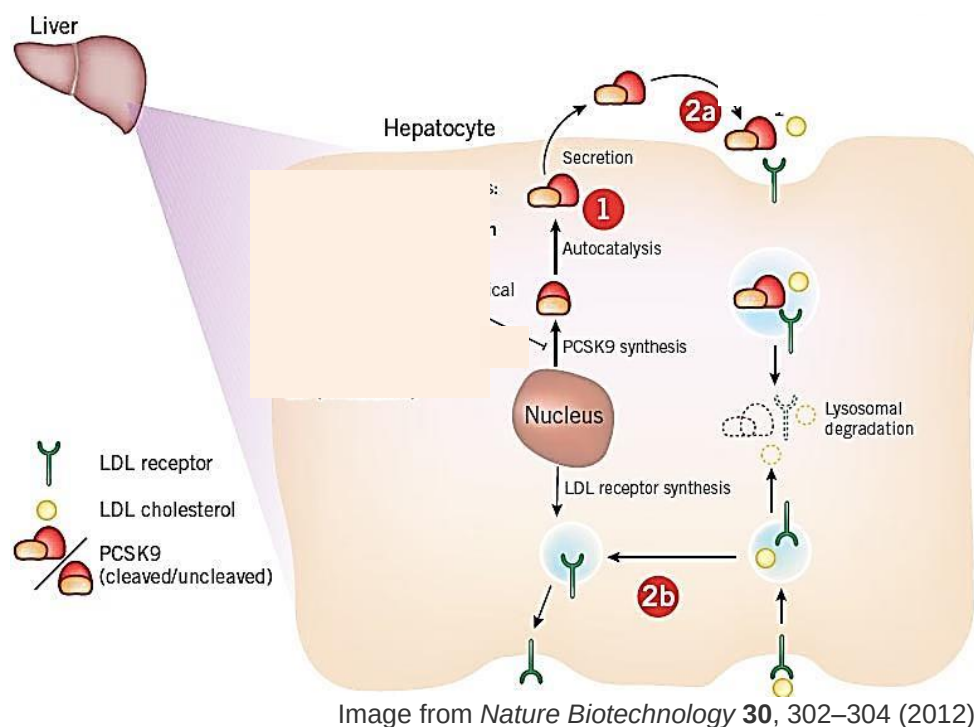


Fig. 4.2

(b) With reference to Fig. 4.2, explain why individuals with increased *PCSK9* production will have elevated levels of LDL cholesterol in the blood. [4]

1. **Gain of function mutation** leading to increased *PCSK9* precursor production in the liver cell;
2. **autocatalytic cleavage** of *PCSK9* precursor to form functional *PCSK9*;
3. before they are **secreted (reject: excreted) out of the liver cell;**
4. Circulating *PCSK9* when complexed with LDL cholesterol that binds to membrane-bound LDL receptor, the complex is internalised / taken into the cell by **receptor-mediated endocytosis** in a clathrin coated vesicle;
5. *PCSK9* directs LDL cholesterol-LDL receptor complex to the **lysosome where it is degraded;**

OR

Vesicle containing *PCSK9* + LDL cholesterol-LDL receptor complex **fuses with membrane of lysosome, contents will be degraded inside lysosome / by lysosome enzymes;**

6. **Fewer LDL receptors are recycled** to the liver cell surface membrane, **limiting the removal** of LDL cholesterol from the circulation, ∴ leading to elevated level / higher than normal level of LDL cholesterol in the blood;
[Point 6 must be mentioned to gain full credit, max 4]

(c) Using the information provided in Fig. 4.2, suggest how drug therapy can be used to inhibit the action of *PCSK9*, so that cholesterol levels can be lowered in patients with familial hypercholesterolaemia. [2]

- **Inhibiting gene expression of PCSK9:** by designing small interfering RNA (siRNA) which binds by complementary base pairing to PCSK9 mRNA, inhibiting PCSK9 translation;
- **Inhibiting gene expression of PCSK9:** by designing antisense oligonucleotides which binds by complementary base pairing to PCSK9 mRNA, inhibiting PCSK9 translation;
- **Inhibiting secretion of PCSK9 into blood circulation:** by designing inhibitor that bind to PCSK9, thereby preventing autocatalysis to form functional PCSK9;
- **Inhibiting action of secreted / circulating PCSK9:** by designing monoclonal antibodies / inhibitors that complex with circulating PCSK9, inhibiting its interaction with LDL receptor;

[Any 2]

[Total: 9]

- 5 The capsules of shepherd's purse occur in two different shapes, i.e., triangular and top-shaped. When a pure breed plant with triangular capsule is crossed with another pure breed plant having top-shaped capsule, all F_1 have only triangular character.

The F_1 offspring were selfed and the following F_2 results were obtained:

1468 triangular-shaped capsules
87 top-shaped capsules

- (a) Using suitable symbols, draw a genetic diagram to explain the results of the F_2 generation. [4]

A – dominant allele for Triangular-shaped capsule
a – recessive allele for top-shaped capsule

B – dominant allele for Triangular-shaped capsule
b – recessive allele for top-shaped capsule

Parental phenotype: Triangular-shaped x Top-shaped
Parental genotype (2n): AABB x aabb

Gametes (n): AB x ab

F_1 genotype (2n): AaBb

F_1 phenotypic ratio: All triangular-shaped capsules

The F_1 generation were selfed

F_1 phenotype: Triangular-shaped x Triangular-shaped
 F_1 genotype (2n): AaBb x AaBb

Gametes (n): AB, Ab, aB, ab x AB, Ab, aB, ab

F_2 genotype (2n):

♀ \ ♂	AB	Ab	aB	ab
AB	AABB Triangular	AABb Triangular	AaBB Triangular	AaBb Triangular
Ab	AABb Triangular	AAbb Triangular	AaBb Triangular	Aabb Triangular
aB	AaBB Triangular	AaBb Triangular	aaBB Triangular	aaBb Triangular
ab	AaBb Triangular	Aabb Triangular	aaBb Triangular	aabb Top

F_2 phenotype: Triangular-shaped capsule : Top-shaped capsule

F_2 phenotypic ratio: 15 : 1
 F_2 observed numbers: 1468 : 87

(b) Explain the genetic basis of this type of inheritance. [3]

- Either alleles **A or B** codes for triangular-shaped capsule.
- Both alleles **a and b** must be present to code for top-shaped capsule

OR

Genotypes at both loci have to be homozygous recessive to code for top-shaped capsule.

- Allele **A** is **epistatic** over allele **b** and Allele **B** is **epistatic** over allele **a**.
- This is a type of duplicate **dominant epistasis**.

The chi-squared (χ^2) test was performed on the results of the second cross, giving a chi-squared calculated value of 2.73.

Part of the critical values of the chi-squared distribution is shown below.

Degrees of freedom	Probability				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(c) Using the information given thus far,

(i) determine the degree of freedom. [1]

- 1

(ii) state what conclusion may be drawn regarding the observed F_2 data. [2]

- Since the chi squared value of 2.76 is less than the critical value of 3.84 at $p = 0.05$, there is no significant difference bet observed and expected numbers;
- The H_0 hypothesis should not rejected and the difference between observed and expected numbers is due to chance alone;

[Total: 10]

- 6 Fig. 6.1 shows some steps involved in glycolysis and the Krebs cycle. Some ATP are made directly. Hydrogen is also released and this can result in the production of more ATP.

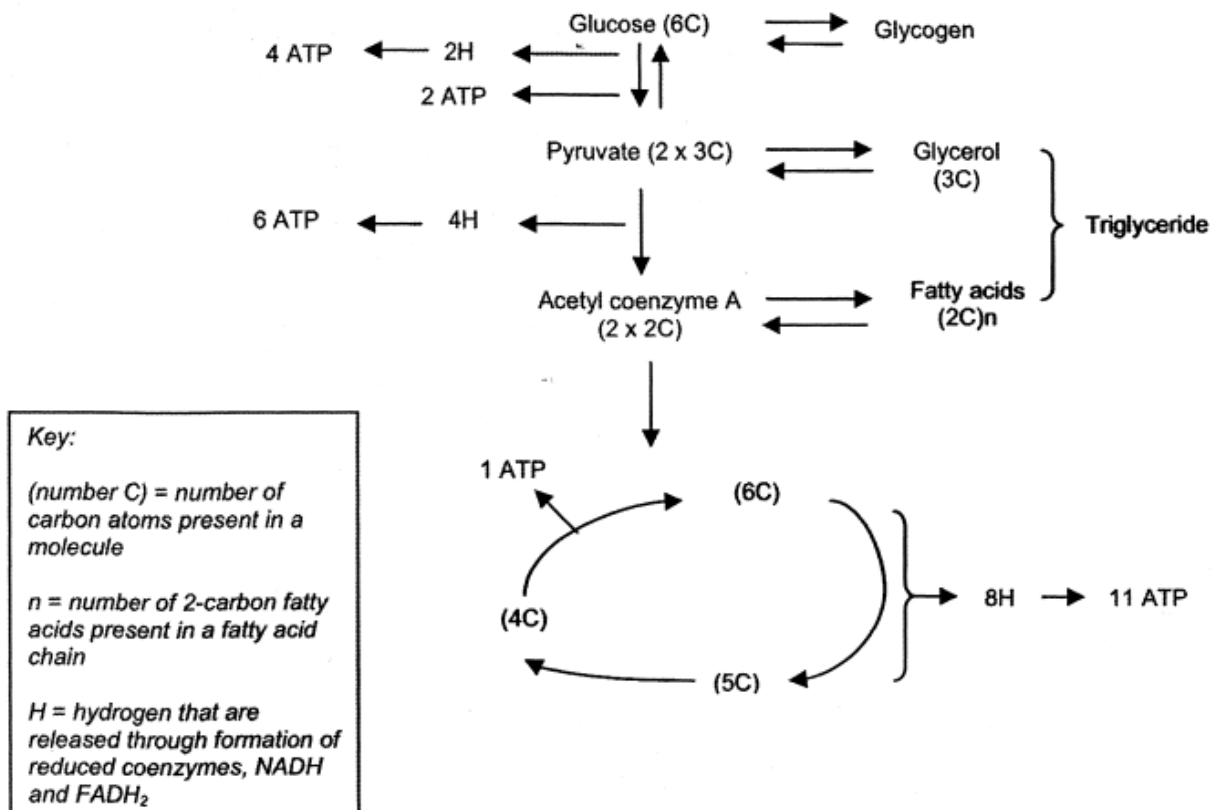


Fig. 6.1

- (a) Describe how the hydrogen released during glycolysis and the Krebs cycle results in the production of ATP. [4]
- Hydrogen released during glycolysis is used to **reduce coenzymes NAD⁺ and FAD⁺**. The NADH and FADH₂ will be reoxidised through transfer of their electrons and protons to the electron transport chain at the inner mitochondrion membrane;
 - Protons are pumped from the mitochondion matrix into the intermembrane space**, generating an **electrochemical gradient**;
 - This electrochemical gradient contains potential energy and the **re-entry of the protons into the mitochondrial matrix** provides the **energy to generate ATP at the ATP synthase**;
 - This process by which ATP is formed as electrons are transferred from NADH and FADH₂ to oxygen via a series of electron carriers is called **oxidative phosphorylation**;

Similar to glycogen, triglycerides can be respired. The first step is to hydrolyse each triglyceride molecule. Each fatty acid formed is broken down into acetyl CoA molecules. The acetyl CoA molecules then enter the Krebs cycle.

(b) Using the information from Fig. 6.1, suggest why fatty acids can only be respired under aerobic conditions.

- Fatty acids enter the mitochondria directly to be respired under aerobic conditions, by being converted to acetyl coA so that it can enter the Krebs cycle. This bypasses the process of glycolysis in the cytosol.

OR

- Glycerol is converted directly into pyruvate in the cytosol. Pyruvate formed then enters the mitochondria to be respired under aerobic conditions, by first undergoing the link reaction where oxidative decarboxylation occurs to form acetyl CoA which then enters Krebs cycle. This bypasses glycolysis.

AND

- In order to be completely oxidised, oxygen is needed as the final electron acceptor in the process of oxidative phosphorylation in order for electron carriers and reduced coenzymes to be reoxidised.

[2]

- (c) In anaerobic respiration in muscle tissues, the conversion of glucose to lactic acid can be represented as a four-stage process shown in Fig. 6.2.

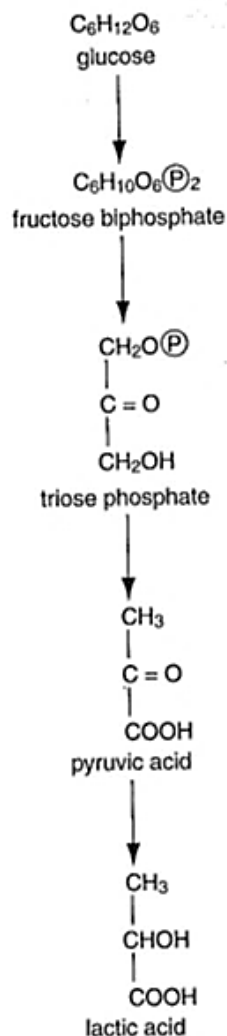


Fig. 6.2

- (i) Explain whether there is any net change in the concentration of [2] coenzymes in the process.
- There is no net change in concentration of NAD^+ and NADH . The NADH formed from oxidation of triose phosphate / glyceraldehyde-3-phosphate to pyruvate is used in the reduction of pyruvate to lactic acid.
 - Similarly, the NAD^+ regenerated during the reduction of pyruvate to lactic acid is used for the oxidation of triose phosphate/ glyceraldehyde-3-phosphate

(ii) If one glucose is converted to two lactic acid molecules, state and explain for the net synthesis of ATP. [2]

- There is a net synthesis of **two ATP molecules**. **Two ATP molecules** are used for phosphorylating glucose to fructose biphosphate making it more chemically active during the energy investment phase.
- **Substrate level phosphorylation** occurs during the energy payoff phase from triose phosphate to pyruvate, to form **four ATP molecules**, transferring the phosphate groups to ADP. Lactate fermentation does not yield ATP molecules.

[Total: 10]

7 Fig. 7.1 shows the plant cell signalling mechanism, which is comparatively less well-studied than that of the animal cell.

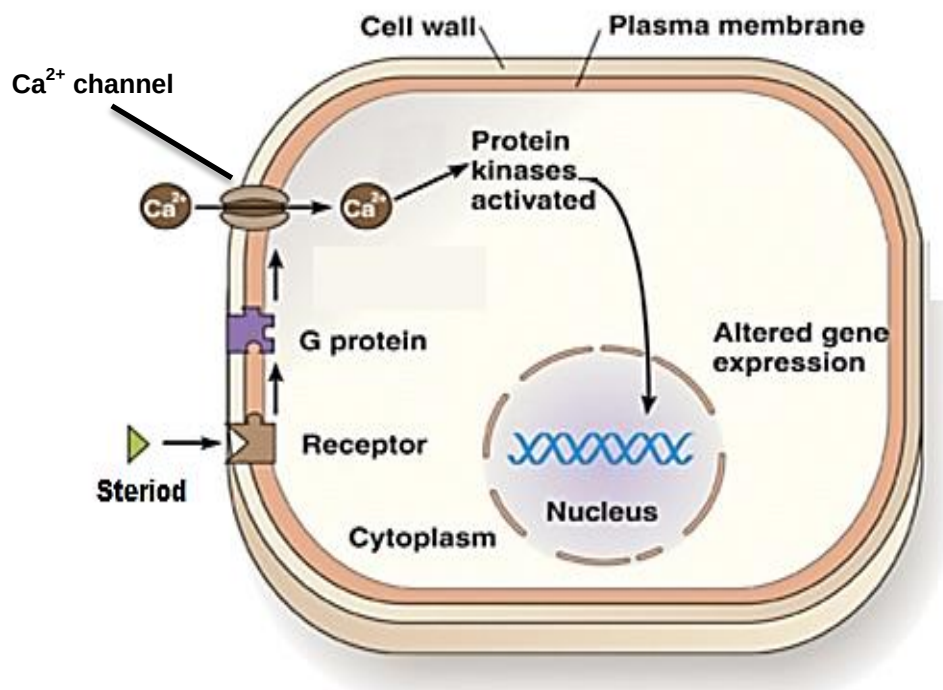


Fig. 7.1

(a) (i) State the main stages of *cell signalling*. [2]

1. Ligand-receptor interaction
/ Signal (molecule preferred) detected at the cell's surface;
2. Signal transduction
/ transmitted into the cell's interior;
3. Cellular response
/ changes in gene expression / cell behaviour;

(ii) With reference to Fig. 7.1, describe the signal transduction process within the plant cell upon steroid binding. [4]

- (Binding of steroid to binding site on extracellular side of receptor triggers) conformational changes which activates the receptor, exposing the cytoplasmic (G protein binding) domain of the receptor;
- G-protein binds and exchanges GDP for GTP and G-protein is activated;
- G-protein dissociates from the receptor, then diffuses along the cell membrane;
- (Binds to and) open the Ca^{2+} channel which allows the influx of Ca^{2+} ions (into the cell);
- Ca^{2+} ions acts as a second messenger by activating protein kinases, which activate proteins via phosphorylation;
- ref. signal amplification / e.g. each Ca^{2+} ion activate a large number of protein kinase thus amplifying the signal;

(b) Explain how dysregulation in cell signalling contributes to cancer progression. [3]

- Constant activation of cell signalling pathway results in activation of the proteins and transcription and translation of the growth factors;
- Cell able to proceed with cell cycle due overexpression of proteins, leading to uncontrolled cell division;
- Formation of tumours, cells able to accumulate mutations, leading to gain of function of proto oncogenes and loss of function of tumour suppressor genes;

(c) Suggest why cancer in plants are often less consequential as compared to cancer in animals. [1]

- Tumor would not metastasize. That's because plant cells are typically locked in place by a matrix of rigid cell walls, so they can't migrate;
- Plants also have the benefit of lacking any vital organs;

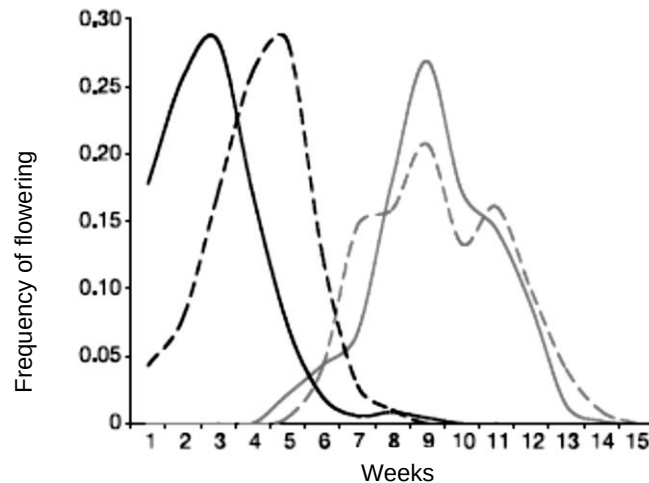
[Max 1]

[Total: 10]

- 8 Lord Howe Island is a small landmass of less than 15 km², 580 km off the eastern coast of Australia.

Two closely related species of palm, *Howea forsteriana* and *Howea belmoreana*, are strikingly different, yet they co-exist in many parts of the island. Both are wind pollinated and diploid ($2n = 32$). Hybrids are uncommon. The two species have both female and male flowers on each individual tree. It is estimated that the two *Howea* species diverged less than one million years ago.

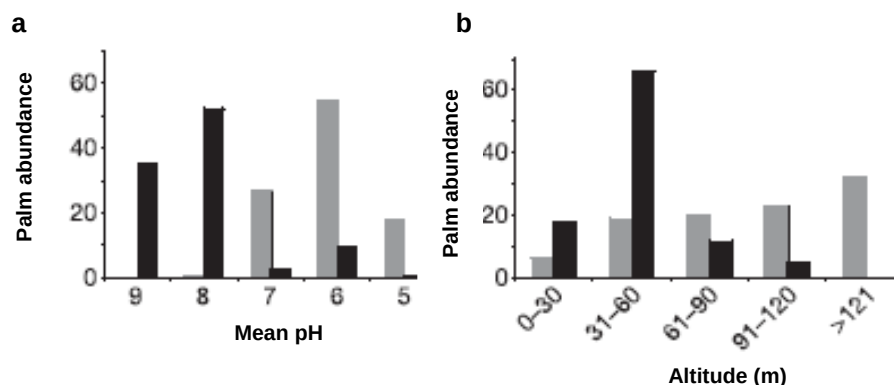
Fig. 8.1 and Fig. 8.2 show the data that have been collected concerning the frequency of flowering and the distribution of the two species respectively.



Flowering frequency for each *Howea* species. *H. belmoreana* is shown in grey, *H. forsteriana* is shown in black, with male flowering (solid line) and female flowering (dotted line) phases.

Image from *Nature* **441**, 210-213 (2006)

Fig. 8.1



Distribution of *H. forsteriana* (black) and *H. belmoreana* (grey), according to mean pH per quadrat in (a) and altitude in (b). Number of palm trees; *H. forsteriana* is shown in black ($n = 1,677$) and *H. belmoreana* in grey ($n = 4,542$). **a, b**, Distribution according to the mean pH of the soil **(a)** in each quadrat and elevation **(b)**. *H. forsteriana* is found in sites at lower elevation and higher soil pH than *H. belmoreana*, and is largely restricted to calcareous soil. However, both species coexist over a large range of elevations and soil pH.

Image from *Nature* **441**, 210-213 (2006)

Fig. 8.2

- (a) With reference to the data in Fig. 8.1 and Fig. 8.2, explain how sympatric speciation may have occurred in *Howea*. [5]

- The differences in flowering times and habitats arose in subpopulations /populations in the ancestral *Howea* species;

2 main reproductive isolating mechanisms (RIMs) appear to be operating:

1. Habitat isolation

The two palm species are more abundant in different habitats.

- *Howea forsteriana* is more abundant in soils with a pH of 8–9 OR in calcareous soil, while *H. belmoreana* is more abundant in soils with a pH of 5-7;
- *H. forsteriana* is found mostly **at low altitudes with a maximum frequency / greater abundance in the 31–60 m range**, while *H. belmoreana* is found at all altitudes but is more common at higher altitudes (> 121 m);

2. Temporal isolation

Flowering periods of the two species differ – when males of one species shed pollen, female parts of the other species are not receptive and interspecific pollination is less likely.

- *H. forsteriana* flowers during **weeks 1–10** of the flowering period (**with a peak at Week 3 for males and Week 5 for females** and **some overlap** between the two sexes);
- *H. belmoreana* flowers in **weeks 5–14** (with **overlap between the two sexes and a peak during Week 9 for both sexes**);
- This difference in flowering times and difference in distribution in the different soil types / altitudes would result in the reproductive / physiological isolation, as only those with the **similar flowering times and living in soils with overlapping pH/ altitudes**, would be able to fertilise one another;
- and hence a disruption of gene flow occurred between the ancestral populations.

[Max 5]

- (b) Discuss the reasons why the speciation event between *H. forsteriana* and *H. belmoreana* is more likely to be sympatric than allopatric. [3]

- Size of island small (15 km²), so geographic isolation not likely on such a small land area; allopatric speciation would require some geographic isolating mechanism that is not feasible on a small island.
- Isolation of island from other landmasses (580 km from Australia) makes it unlikely that one species has evolved elsewhere and arrived on the island by dispersal.
- Wind pollination – pollen distributed by the wind can be blown considerable distances; likely that interspecific pollination would occur on a small island.
- Habitat isolation could have separated two populations and helped prevent gene flow.
- Temporal isolation could have separated two populations and helped prevent gene flow; could have arisen by a simple mutation changing the flowering period of an individual which resulted in that individual producing a new species.

[max 3]

- (c) *Howea belmoreana* goes by the common names, curly palm, kentia palm and Belmore sentry palm while *Howea forsteriana* goes by the names thatch palm and kentia palm.

Explain why the scientific name is better than the common names.

[2]

1. It is unambiguous and precise with one unique name for one species;
2. unlike the common name "Kentia palm" which is used to refer to both species or multiple names referring to one species (evidence);

Or

3. The binomial nomenclature is a systematic naming system that allows one to categorise/group similar organisms together involving both the species name and genus name (evidence);
4. while the common names do not reveal their position in the plant kingdom nor attempts to categorise them;

🚩 **1 and 2 (link) or 3 and 4 (link). If 1 and 3 or 4 allow only 1 mark (unlink)**

[Total: 10]

Section B

Answer any **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)**, etc., as indicated in the question.

9 (a) Compare repressible and inducible systems of gene regulation.

[7]

Type of negative operons	Inducible operon	Repressible operon
Named example	<i>Lac</i> operon	<i>Trp</i> operon
Default operon expression	Switched off	Switched on
Type of metabolic pathway	Catabolic, e.g. breakdown of lactose	Anabolic, e.g. synthesis of tryptophan
Usual state of repressor	Active and is bound to the operator, preventing the initiation of transcription by blocking RNA polymerase from binding to the promoter	Inactive and does not bind to the operator, allowing transcription initiation when RNA polymerase binds to the promoter
Molecule bound to repressor (to alter its usual state)	Inducer binds to repressor to inactivate it, by changing it to its inactive conformation. The repressor can no longer bind to the operator, allowing transcription initiation when RNA polymerase binds to the promoter.	Co-repressor binds to the repressor to activate it by changing it to its active conformation. The repressor can now bind to the operator, preventing the initiation of transcription by blocking RNA polymerase from binding to the promoter.
Conditions under which transcription of structural genes occurs	Availability of substrates	Insufficient amounts of end product
Significance	Ensures that enzymes are only synthesised when required, thus conserving energy and resources	Ensures that resources are not used to synthesise products already present in sufficient quantities, thus conserving available energy and resources

- (b) Explain, with appropriate examples, how recessive alleles may be preserved in a natural population.

[8]

Diploidy

1. Recessive alleles that contribute negatively to an organism's fitness can be hidden in heterozygotes, which continue to propagate them / the diploid nature of most eukaryotes hides recessive alleles that are less favourable, from selection in the form of heterozygotes;
2. Dominant alleles mask the effects of recessive alleles, protecting them from natural selection;
3. Only **homozygous recessive** individuals may be **selected against**, and such individuals are rarer when the frequency of recessive alleles is smaller / the latent variation is exposed to selection only when each parent carries the same recessive allele and combines the two copies in their zygote / occurs in the homozygous recessive state;
4. Maintains a huge pool of alleles which might not be favourable under present circumstances but which could be beneficial when the environment changes.
5. any relevant e.g.;

Heterozygote advantage

5. Occurs when individuals who are heterozygous at a particular gene locus have greater survivorship and reproductive success than any type of homozygote;

Example:

6. In malaria-prone regions, individuals who are heterozygous at the gene locus coding for β -globin of haemoglobin have a greater fitness / Heterozygotes are resistant to malaria.
7. These heterozygotes thus have the heterozygote advantage in malaria prone areas;
8. Homozygotes either die due to malaria (homozygous dominant) or due to sickle cell disease (homozygous recessive);

Inbreeding

9. In a small population, matings between relatives are common. This **inbreeding** may lower the population's ability to survive and reproduce, a phenomenon called **inbreeding depression**.
10. **Natural selection weeds deleterious alleles out of a population over time** because when the dominant deleterious alleles are expressed, they lower the carrier's fitness, and fewer copies wind up in the next generation.
11. But recessive deleterious alleles are "hidden" from natural selection by their dominant non-deleterious counterparts (**heterozygote protection**).
12. An individual carrying a single recessive deleterious allele will be healthy and can easily pass the deleterious allele into the next generation. When the population is large, this is generally not a problem—the population may carry many recessive deleterious alleles, but they are rarely expressed.
13. However, **when the population becomes small, close relatives end up mating with one another, and those relatives likely carry the same recessive deleterious alleles**. When the relatives mate, the **offspring may inherit two**

copies of the same recessive deleterious allele and suffer the consequences of expressing the deleterious allele.

14. For example, a population of 40 adders (*Vipera berus*) experienced inbreeding depression when farming activities in Sweden isolated them from other adder populations.
15. Higher proportions of stillborn and deformed offspring were born in the isolated population than in the larger populations.

[max 8]

(c) Explain how the structures of centromeres are linked to their roles.

[5]

Feature	Discussion of roles
Centromeres <u>are constricted</u> region of a chromosome Compose of alpha satellite <u>DNA</u> sequences and are AT-rich and consist of a large number of repeats [1] Flanking → centric heterochromatin, which contains DNA sequences, composed of different types of repeats.	They are essential in nuclear division→ holds sister chromatids [1] The heterochromatin state of centromeres appear to be essential for the recruitment of cohesin complexes that mediate sister chromatid adhesion after DNA replication [1]
Kinetochores form <u>on the region</u> of a satellite DNA of centromeres and made up of proteins. [1]	For spindle fibres to attach. [1] During mitosis, the chromosomes must align first before separation The centromere that is shared by sister chromatids must divide so that the chromatids can migrate to opposite poles of the cell. Therefore the centromere is an important component of chromosome structure and segregation /responsible for the accurate segregation of the replicated chromosome during mitosis and meiosis. [1] Without a properly functioning centromere, improper chromosomal alignment and segregation will result, leading to aneuploidy and conditions such as Down's syndrome.

[Total: 20]

10 (a) Compare binary fission with mitosis.

[7]

Similarities:

1. All cells have DNA that must be copied, segregated and distributed to new cells in an organized fashion. Both interphase (prior to mitosis) and binary fission accomplish this.
2. The process begins when new cells are needed. The cell grows larger, makes copies of all components, and then replicates the DNA.
3. Replicated DNA is allocated equally to the new cells and the cell splits in half. Cell membrane invaginates (fold inwards) in binary fission while in cytokinesis in animal cells, a cleavage furrow is formed and eventually separates the two daughter cells.
4. Both processes form two genetically identical cells.

Differences:

5. Binary fission occurs among prokaryotes (cells that do not contain a nucleus) while Mitosis occurs among eukaryotes (cells that have a nucleus).
6. Binary Fission → DNA is circular and has no free ends, an interlocking structure made up of 2 daughter molecules formed. In mitosis → linear DNA so no such interlocking structure formed
7. Binary Fission → Topoisomerase is needed to cut, separate and reseal two DNA molecules. Mitosis → linear DNA so no such interlocking structure formed.
8. Binary fission - no disintegration of nuclear envelope due to absence of a true nucleus while in mitosis, disintegration of nuclear envelope occurs during prophase.
9. Binary fission does not include spindle formation (mitotic apparatus), rather each duplicated DNA molecules eventually attached to the plasma membrane. In mitosis, spindle fibres formed to attach to the centromere where sister chromatids are joined together to separate the sister chromatids during anaphase.
10. Binary fission does not have the four distinct cellular phases while mitosis is made up of Prophase, Metaphase, Anaphase and Telophase.

(b) Explain the control mechanisms involved in the expression of the eukaryotic genes at the translational and post-translational level. [8]

Regulation at the initiation of translational stage:

1. Availability of eukaryotic initiation factors (eIFs)
 - eIFs are additional proteins that promote the proper association of ribosomes to mRNA and are required for protein synthesis to take place.
2. Binding of translational repressors to the 5' cap or 5' untranslated region (UTR) (leader)
 - This prevents binding of small ribosomal subunit to 5' cap
3. Binding of translational repressors to the poly(A) tail
 - Prevents eukaryotic initiation factors (eIFs) on the poly(A) tail from binding to those on the 5' cap, thus preventing the small ribosomal subunit from binding to the 5' cap

Regulation at the elongation of translational stage:

1. Rate of protein synthesis can also be regulated by the availability of amino acids and regulatory proteins required for translation.

Other points also accepted:

- The presence of polysomes / polyribosomes can speed up the rate of translation for a particular mRNA.
- Rate of mRNA degradation will affect the level of translation. If mRNA are degraded, translation will not take place.
- Rate of mRNA processing will affect the rate of mature mRNA which will affect the rate of translation.

Regulation at post-translational stage:

4. Chemical modification by addition of chemical groups to the polypeptide chain:
 - a. Phosphorylation – addition of phosphate group (can activate or deactivate protein, depending on type of protein)
 - b. Glycosylation – addition of one or more sugar monomers to form glycoproteins
 - c. Ubiquitination – addition of a ubiquitin (a small protein) which identifies the protein for degradation
5. Modification of the sequence of amino acids in the polypeptide chain:
 - Removal of inhibitory portions of the polypeptide chain by protease, for proteins which are synthesized as a larger protein precursor before it can be activated.
6. Protein degradation
 - The longer a protein stays in the cell before it is degraded, the longer it can carry out its functions, thus the expression of the gene that codes for this protein is higher.

(c) Explain the different consequences of chromosomal mutation.

[5]

Definition of mutation:

- Change in the amount, arrangement or structure of the DNA of an organism.
- Produces a change in the genotype and may result in the change in appearance of a characteristic in a population.

Definition of Chromosomal mutations:

- Result of changes in the number and structure of chromosomes.
- **Chromosome mutation:** Changes in chromosome number;
- These changes may involve the loss or gain of single chromosomes, a condition called **aneuploidy**;
- Or the increase in entire haploid sets of chromosomes, a condition called **euploidy (polyploidy)**;

Aneuploidy

- Loss of gain or single chromosomes e.g. (n+l), (2n+l), (n-1), (2n-1);
- Result from the failure of a pair, or pairs of homologous chromosomes to separate during anaphase I of meiosis. This is known as non-disjunction;
- **Any named example:**
 1. **Down's syndrome (Mongolism)**
 - Presence of an extra chromosome no. 21 in the cells.
 - Symptoms – mental retardation, short stocky body, thick neck, characteristic folds of

- skin over the inner corner of the eye.
- Related to the age of the mother's egg cells.

2. Klinefelter's syndrome (XXY)

- Failure of X chromosome to separate during oogenesis in female.
- Symptoms – male with female characteristic, sterile, testes very small, little facial hair, breasts may develop, low intelligence.

3. Turner's syndrome (XO)

- Sterile female.
- Lacking normal secondary sexual characteristics.

4. XXX

- Female, normal appearance
- Fertile but mentally retarded;

5. XYY

- Male, variable intelligence, may possess psychopathic traits or tendency for petty criminal acts;
- **Euploidy (Polyploidy):** Cells contain multiples of the haploid number of chromosomes i.e. polyploids $2n$, $3n$,.....;
- More common in plants than animals;
- Associated with hybrid vigour;
- **Two forms:**
 1. **Autopolyploids:** An individual with more than 2 sets of chromosomes all derived from the same species;
 - Can be induced by using colchicine;
 - Autopolyploids can be fertile if they have an even number of chromosome sets;
 2. **Allopolyploids:** An individual with increased chromosome number derived from different haploid sets;
 - Many inter-species hybrids are sterile because chromosomes do not pair up at meiosis, resulting in a diploid hybrid (sterile);
 - If chromosomes are doubled, each member has another to pair up with and meiosis proceeds normally -- fertile, allopolyploid;
- **Chromosome mutation: Changes in structure:**
 1. Deletion: removes a chromosomal segment;
 2. Inversion: reverses a segment within a chromosome;
 3. Translocation: moves a segment from one chromosome to another, non-homologous one;
 4. Duplication: repeats a segment;

Any named example:

1. Cri-du-chat syndrome:

- deletion in the short arm of chromosome 5;
- child is physically and mentally retarded;

2. Chronic myelogenous leukaemia (CML):

- Translocation
- Most of the chromosome 22 has been translocated onto the long arm of chromosome 9;
- In addition, the small distal portion of chromosome 9 is translocated to chromosome 22;

- Caused increased cell proliferation, reduced apoptosis, leading to cancer;

3. Acute myelogenous leukaemia (AML):

- Chromosomal inversion on chromosome 1;
- Results in cancer;

4. Charcot-Marie-Tooth syndrome (CMT):

- In one form, chromosomal duplication on chromosome 17;
- High gene dosage of myelin sheath protein resulting in abnormal structure and function of the myelin sheath;
- Symptoms: weakness of lower foot, loss of balance, poor motor skills and muscle atrophy;

[Total: 20]

END OF PAPER